

LAB REPORT OCR SYSTEM - USER MANUAL

Step-by-Step Guide for Installation, Operation, Configuration, and Troubleshooting

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1. Introduction

Common Error: 'TesseractNotFoundError'

Fix: Ensure Tesseract is installed on your OS path, or modify your environment path settings.

For low OCR accuracy, binarize crop images using the preprocessing configs or sharpen image contrasts.

2. System Requirements

Ensure your local system meets the following specifications:

- CPU: Intel Core i5 (4th Gen) / AMD Ryzen 5 or higher
- RAM: 8 GB minimum (16 GB recommended)
- OS: Windows 10/11, macOS Big Sur+, Ubuntu 20.04+

3. Installation Guide

1. Install Python 3.11 or later.
2. Install Tesseract OCR:
 - macOS: brew install tesseract
 - Ubuntu: sudo apt-get install tesseract-ocr
3. In terminal, navigate to the folder and run:

```
python -m venv venv
```

```
source venv/bin/activate
```

```
python -m pip install -r requirements.txt
```

4. How to Run the App

To start the application locally, run:

```
streamlit run app/app.py
```

Then open your web browser and go to <http://localhost:8501>

5. Operating the System

Upload images or PDFs using the uploader, choose your confidence threshold in the sidebar config, select Tesseract or EasyOCR, and hit run. Results are displayed as an editable table.

6. Troubleshooting & FAQs

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